Experiment 1

Comparative Study of Low-Code Development Versus Traditional Application Development Approaches

Aim

To compare low-code development and traditional application development approaches based on development speed, ease of use, customization, cost, maintenance, scalability, and overall suitability for different types of software projects.

Theory

Low-code development platforms provide graphical user interfaces, drag-and-drop components, and pre-built templates that allow developers to build applications with minimal manual coding. They reduce development time and enable both professional developers and citizen developers to create applications quickly.

Traditional development involves writing source code using languages such as Java, Python, C#, or JavaScript. Developers have complete control over design, architecture, and functionality, at the cost of more programming knowledge and development time.

Procedure

1. Study the concepts of low-code development and traditional application development.
2. Select a low-code platform and review its features, including drag-and-drop interface, templates, and automation tools.
3. Study the traditional software development process using programming languages and frameworks.
4. Compare both approaches based on development time, coding effort, customization, scalability, cost, maintenance, and deployment.
5. Record the observations in a comparison table.
6. Analyze the advantages and limitations of each approach.
7. Draw conclusions regarding the most suitable development approach for different application requirements.

Comparison Table

Parameter	Low-Code Development	Traditional Development
Development Time	Faster	Slower
Coding Requirement	Minimal	Extensive
Learning Curve	Easy	Moderate to Difficult
Customization	Limited	Highly Flexible
Cost	Lower for small projects	Higher initially
Scalability	Moderate	High
Maintenance	Easier	Requires skilled developers

Parameter	Low-Code Development	Traditional Development
Best Suited For	Simple to medium business applications	Complex enterprise applications

Result

The comparative study shows that low-code development significantly reduces development time and coding effort, making it suitable for rapid application development and business applications. Traditional application development, while requiring greater time and programming expertise, provides superior flexibility, scalability, performance, and customization, making it the preferred choice for large-scale and complex software systems.